Fiscal Year:	FY 2009	**Task Last Updated:**	FY 07/03/2009
PI Name:	Bacher, Jeff Ph.D.		
Project Title:	A Novel Biodosimetry Method		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	jeff.bacher@promega.com	**Fax:**	FY 608-273-6989
PI Organization Type:	INDUSTRY	**Phone:**	608-277-2608
Organization Name:	Promega Corporation		
PI Address 1:	Genetic Analysis		
PI Address 2:	2800 Woods Hollow Road		
PI Web Page:			
City:	Madison	**State:**	WI
Zip Code:	53711-5399	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2007 Space Radiation NNJ07ZSA001N
Start Date:	09/01/2007	**End Date:**	08/31/2011
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
Contact Monitor:	Cucinott1a, Francis	**Contact Phone:**	281-483-0968
Contact Email:	noaccess@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: Received NCE through 8/31/2011, per C. Guidry/JSC (08/2010)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Bailey, Susan (Colorado State University) Halberg, Richard (University of Wisconsin)		
Grant/Contract No.:	NNX07AQ02G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Exposure of astronauts to space radiation during extended space missions may cause serious health problems. Accurate methods for measuring the biological effects of radiation exposure are, therefore, critical for estimating an individual's health risks. Biodosimetry measurements reflect variation in radiation sensitivity and consequently result in highly individualized estimates of dose and risk. Our novel biodosimetry approach is based on the hypothesis that non-coding repetitive DNA sequences are sensitive to radiation-induced mutations and that these mutations are not harmful to a cell. Therefore, mutations in non-coding repetitive DNA sequences can accumulate and provide a stable molecular record of genetic damage that can be used to determine cumulative radiation exposure and health risk. In our previous NASA grant, we demonstrated the feasibility of using radiation-induced mutations in non-coding repetitive DNA sequences to estimate radiation dose. Our initial data indicate that radiation-induced mutations in non-coding repetitive DNA markers are stable over time and additive over multiple exposures. In this successor proposal, we plan to extend our previous work by developing optimized multiplex marker panels for human and mouse biodosimetry, validate our approach by comparing our assay to current gold standard cytological methods and then utilize the novel system to assess risks from space radiation and improve our understanding of how these risks are affected by variations in dose rate, dose fractionation and genome stability. The main contribution of the proposed research to manned space exploration is the validation of a novel biodosimetry method for estimating dose and risks from exposure to space radiation. Completion of this research should provide new insights into the effects of space radiation on DNA mutagenesis and establishes panels of human and mouse biomarkers with broad utility for future studies in radiation biology, toxicology and cancer research.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Mutational load profiling, through analysis of mutations in tandem DNA repeat sequences, is a simple, non-invasive and generalized approach for monitoring an individual's cumulative record of mutations that may be useful for determining health risks and effectiveness of countermeasures for astronauts or other individuals exposed to ionizing radiation or chemical mutagens. Biomarkers identified in this study are also sensitive to free radical DNA damage and therefore may be useful markers for detection of cancer and other degenerative diseases in which oxidative stress is involved. Completion of this research should provide new insights into the effects of space radiation on DNA mutagenesis and establishes panels of human and mouse biomarkers with broad utility for future studies in radiation biology, toxicology and cancer research.
Task Progress:	Irradiation and sample collection began in the first year and has been successfully completed for all but two of our experiments, which are scheduled for fall 2009. In the 1st year we focused on identifying new mouse and human microsatellite markers and constructed mouse and human multiplexes assay for small-pool PCR (SP-PCR). In the 2nd year we have improved our SP-PCR assay to dramatically increase throughput and accuracy while reducing costs. Mutational analysis of samples began in the later part of the first year and became the primary focus in the later part of the 2nd year. We have been able to successfully culture primary buccal cells and therefore will be using primary human blood and buccal cells for dose response studies instead of transformed cell lines. Human research protocols these new experiments have been approved by our institute and are under review by Brookhaven National Laboratory. A brief description of each planned experiment and current status is given below. Experiment 1. Biomarker discovery: Suitable microsatellite repeats in mouse and human genomes were identified and screened for sensitivity to radiation-induced mutations. We identified over 300 new microsatellite markers and constructed mouse and human multiplexes assay for small-pool PCR (SP-PCR). Status: completed. Experiment 2. Dose Response in mice: Mice were exposed to various doses of HZE iron particles, protons or gamma rays, allowed to recover for 3 days, then samples were collected for later mutation testing. Status: irradiations and sample collection completed. Experiment 3. Dose Response in human cells: Cultured primary human buccal and blood cells will be irradiated at NSRL-09C and analyzed for mutations using SP-PCR. IRB protocols are under review. Experiment 4. Stability: Blood and buccal samples from B6 mice were be exposed to iron ions, protons or gamma rays and analyzed for mutations after 3 days, 3 mo, 12 mo and 18 mo. Irradiations have been completed on all mice and samples collected for all time points except 18 months. Analysis of samples is ongoing. Experiment 5. Mutations in coding repeats: Irradiated mismatch repair deficient mice containing the SupFG1 mutation reporter were screened for mutations in (C)8 and (G)7 coding repeats of the SupFG1 gene. DNA sequencing revealed that 97% of the SupFG1 mutations were insertions or deletions in the (G)7 or the (C)8 mononucleotide repeats. Mutations in these short repeats exited a dose response that was highly correlated (but about 10-fold lower) with mutations in longer non-coding repeats. Status: completed. Experiment 6. Dose rate effects: B6 mice were irradiated with 1 Gy iron, protons and gamma rays at dose rates of 0, 10, 20, 50, 100 and 150 cGy/minute, then analyzed for mutations by SP-PCR. Preliminary results indicate that mutation frequency increases linearly in mouse blood cells with increasing dose rates up to 50 cGy/min iron ions. This trend did not occur with protons or gamma rays, but we will need to analyze more samples to confirm this. Status: analysis ongoing. Experiment 7. Dose- fractionation: B6 mice were exposed to 0.5 Gy of iron, protons or gamma rays at 0.5 Gy/m followed by another 0.5 Gy dose 3 days later. Mutation frequency after a single 1 Gy dose will be compared to 2x 0.5 Gy dose. Status: irradiations and sample collection complete. Experiment 8. Adaptive response: B6 mice were exposed to different combinations of a 0.1 Gy priming dose followed 24 hr later by a 0.9 Gy challenge dose. Samples were collected 3 days later for mutation analysis by SP-PCR to determine if mutation frequencies are additive, lower than expected or higher than expected. Status: irradiations and sample collection complete. Experiment 9. Aging effects: Young (6 wk) and old (18 mo) B6, CBA/Ca and Balb/c mice will be irradiated and screened for mutations by SP-PCR and chromosomal and telomere aberrations using Giemsa staining and telomere FISH. Status: scheduled for NSRL-09C. Experiment 10. DNA mismatch repair expression: qPCR, IHC, Western Blot and methylation specific PCR assays will be performed on tissues from irradiated mice to determine whether mismatch repair gene expression or methylation

	patterns are altered following radiation exposure. Status: irradiations and sample collection complete. qPCR and IHC assays are in development. Experiment 11. Tissue Effects: Msh2-/- deficient mice were irradiated with 1 Gy iron ions and are being screened for microsatellite mutations in blood, buccal cells, spleen, colon, liver and brain tissues. Mutation frequency differences were observed in different tissues. Msh2+/+ and +/- tissues exhibit similar mutation frequencies. Msh2 deficient tissues had approximately 10-fold higher rates of mutation rates compared to wild type. Status: irradiations and sample collection complete. Analysis ongoing. Experiment 12. LET Effects: Mice were irradiated with iron ions of different LET (300, 600 and 1000 MeV/n) and blood samples will be tested for mutations by SP-PCR. Status: Irradiations and sample collection complete.
Bibliography Type:	Description: (Last Updated: 04/16/2019)
Abstracts for Journals and Proceedings	Bourdeau-Heller J, Steffen L, Leisemann R, Bailey S, Halberg R, Stanhope S, Storts D, Bacher J. "Microsatellite Mutations Induced by High and Low-LET Radiation." Radiation mutagenesis session, Heavy Ions in Therapy and Space Symposium, Cologne, Germany, July 6-10, 2009. Heavy Ions in Therapy and Space Symposium, Cologne, Germany, July 2009. p. 230. http://www.heavyions2009.de/images/pdf/HeavyIons_AbstractBook_B5.pdf , Jul-2009
Abstracts for Journals and Proceedings	Steffen L, Weil M, Ray F, Genik P, Ullrich R, Fallgren C, Story MD, Gillan J, Bouffler S, Bourdeau-Heller J, Leisemann R, Bacher J. "Microsatellite Instability in Radiation-Induced Acute Myeloid Leukemia." Radiation carcinogenesis session, Heavy Ions in Therapy and Space Symposium, Cologne, Germany, July 6-10, 2009. Heavy Ions in Therapy and Space Symposium, Cologne, Germany, July 2009. p. 223. http://www.heavyions2009.de/images/pdf/HeavyIons_AbstractBook_B5.pdf , Jul-2009
Articles in Peer-reviewed Journals	Haines J, Bacher J, Huiskamp R, Meijne E, Mancuso M, Pazzaglia S, Bouffler S. "Microsatellite Instability in Radiation-induced Murine tumours; Influence of Tumour Type and Radiation Quality." International Journal of Radiation Research (in press). Expected publication December 2009. , Jul-2009
Articles in Peer-reviewed Journals	Halberg R, Jesse Waggoner J, Rasmussen K, White A, Clipson L, Bacher J, Sullivan R, Washington M, Pitot H, Petrini J, Albertson D, Dove W. "Long-lived Min Mice Develop Advanced Intestinal Cancers Through A Genetically Conservative Pathway." Cancer Research (in press). Expected publication December 2009. , Jul-2009